6 **(a)** State how a line emission spectrum may be explained on the basis of the existence of discrete electron energy levels in atoms.

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.....[3]

(b) The lowest six discrete energy levels for a hydrogen atom are shown in Fig. 6.1. The diagram is drawn to scale and the ground state is at -13.6 eV.

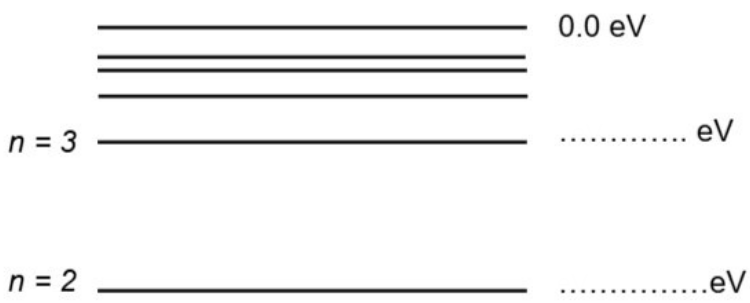


Fig. 6.1 (Drawn to scale)

(i) On Fig. 6.1, label the values for energy at energy levels $n = 2$ and $n = 3$. [2]

(ii) Calculate the wavelength of the light emitted when an electron falls from $n = 3$ energy level to the $n = 2$ energy level.

wavelength = ... m [2]

(iii) Describe one method by which an electron in a gaseous hydrogen atom can be excited from a lower energy level to a higher energy level.

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.....[1]

[Total: 8]